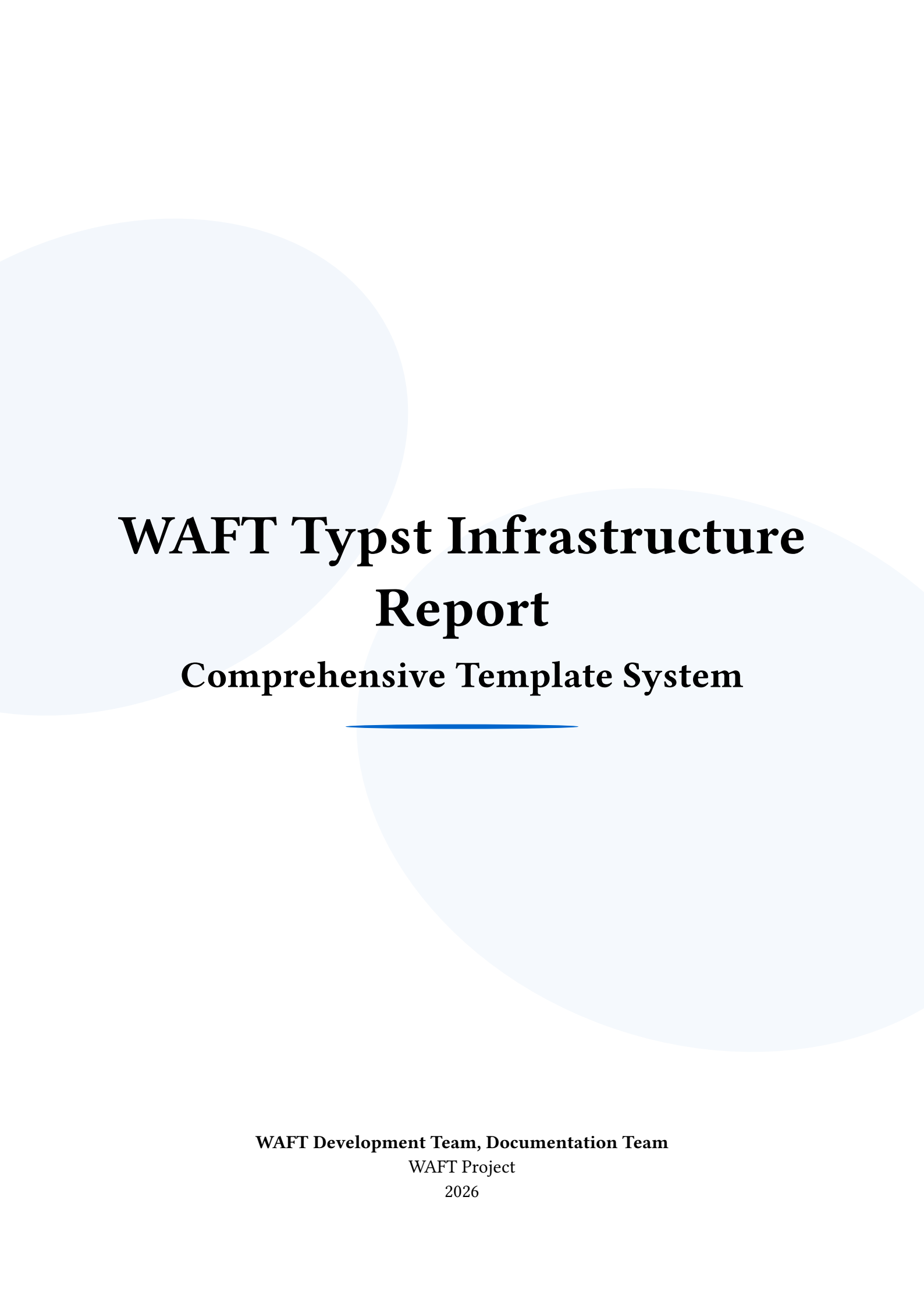




WAF T Typst Infrastructure Report

Comprehensive Template System

WAF T Development Team, Documentation Team
WAF T Project
2026

Contents

1. Executive Summary	2
1.1. Architecture	2
1.2. Security Features	2
1.3. Performance	2
1.4. Template Categories	2
2. Conclusion	2

1. Executive Summary

The Typst infrastructure provides a modern, secure, and efficient system for PDF generation.

1.1. Architecture

The system consists of three main components:

1. **TypstCompiler**: Secure compilation engine with hardening
2. **TypstTemplateRegistry**: Auto-discovery and metadata extraction
3. **Template Wrappers**: Unified API layer for all templates

1.2. Security Features

- Path validation prevents directory traversal attacks
- Content size limits prevent resource exhaustion
- Timeout protection prevents hanging processes
- Subprocess security with `shell=False`

1.3. Performance

Typst compilation is significantly faster than LaTeX:

- Single-pass compilation (vs. LaTeX's multi-pass)
- Typical compilation time: < 1 second
- Efficient resource usage

1.4. Template Categories

The registry includes templates across multiple categories:

- Academic papers (preprint, IEEE, AMS)
- Books and reports
- Letters and newsletters
- Interactive games

2. Conclusion

The Typst infrastructure successfully provides a modern alternative to LaTeX with enhanced security and performance.